Healthcare Analysis

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1. Introduction

This healthcare analysis project leverages a synthetic dataset of over 55,000 patient records to uncover trends in admissions, length of stay, and billing patterns. As a data analyst, I applied SQL for data cleaning and transformation by removing duplicates, standardizing patient names, and calculating stay lengths to prepare the data for analysis. Using Power BI, I created interactive dashboards with custom DAX measures to visualize key healthcare metrics and support data-driven decision-making. This project showcases core data analysis skills including data manipulation, visualization, and business insight generation within the healthcare domain.

2. Dataset Overview

This synthetic healthcare dataset (from Kaggle) contains 15 columns and mimics real patient records. It’s designed for practicing data analysis in a healthcare context. https://www.kaggle.com/datasets/prasad22/healthcare-dataset

Key Features:

- Demographics: Age, Gender, Blood Type
- Medical Info: Condition, Medication, Test Results
- Admission: Dates, Doctor, Hospital, Room

Raw Data

	name character varying (100)	age smallint	gender character varying (10)	blood_type character varying (3)	medical_condition character varying (100)	date_of_admission date	doctor character varying (100)	hospital character varying (150)
1	Bobby JacksOn	30	Male	B-	Cancer	2024-01-31	Matthew Smith	Sons and Miller
2	LesLie TErRy	62	Male	A+	Obesity	2019-08-20	Samantha Davies	Kim Inc
3	DaNnY sMitH	76	Female	A-	Obesity	2022-09-22	Tiffany Mitchell	Cook PLC
4	andrEw waTtS	28	Female	O+	Diabetes	2020-11-18	Kevin Wells	Hernandez Rogers and Vang,
5	adrIENNE bEll	43	Female	AB+	Cancer	2022-09-19	Kathleen Hanna	White-White

insurance_provider character varying (50)	billing_amount numeric (12,2)	room_number integer	admission_type character varying (50)	discharge_date date	medication character varying (100)	test_results character varying (50)
Blue Cross	18856.28	328	Urgent	2024-02-02	Paracetamol	Normal
Medicare	33643.33	265	Emergency	2019-08-26	Ibuprofen	Inconclusive
Aetna	27955.10	205	Emergency	2022-10-07	Aspirin	Normal
Medicare	37909.78	450	Elective	2020-12-18	Ibuprofen	Abnormal
Aetna	14238.32	458	Urgent	2022-10-09	Penicillin	Abnormal

3. Finding Duplicates

```sql
WITH ranked AS (
    SELECT ROW_NUMBER() OVER (
                PARTITION BY name, age, gender, blood_type, medical_condition, date_of_admission,
                             doctor, hospital, insurance_provider, billing_amount, room_number,
                             admission_type, discharge_date, medication, test_results
                ORDER BY ctid
            ) AS rn
    FROM healthcare
)
SELECT COUNT(*) AS duplicate_count
FROM ranked
WHERE rn > 1;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	duplicate_count bigint
1	534

Deleting Duplicates

```
DELETE FROM healthcare
WHERE ctid NOT IN (
    SELECT MIN(ctid)
    FROM healthcare
    GROUP BY
        name, age, gender, blood_type, medical_condition, date_of_admission,
        doctor, hospital, insurance_provider, billing_amount, room_number,
        admission_type, discharge_date, medication, test_results
);
```

Data Output Messages Notifications

```
DELETE 534

Query returned successfully in 1 secs 378 msec.
```

Standardize Spelling in name

Before

	name character varying (100)
1	Bobby JacksOn
2	LesLie TErRy
3	DaNnY sMitH
4	andrEw waTtS
5	adrIENNE bEll

```
UPDATE healthcare SET name = INITCAP(name);
```

After

	name character varying (100)
1	Mary Hunt
2	Mrs. Sabrina Ball
3	Sandra Murphy
4	Angela Herring
5	John Johnson

Cleaned Data

	name character varying (100)	age smallint	gender character varying (10)	blood_type character varying (3)	medical_condition character varying (100)	date_of_admission date	doctor character varying (100)	hospital character varying (150)
1	Mary Hunt	44	Male	AB-	Hypertension	2020-08-31	Stephanie Barnett	Guzman LLC
2	Mrs. Sabrina Ball	63	Male	AB-	Asthma	2020-11-11	Tina Griffin	Burch-Marks
3	Sandra Murphy	84	Male	A-	Asthma	2019-11-29	Angela Clayton	Bruce-Thomas
4	Angela Herring	52	Female	A+	Asthma	2023-11-07	Alicia Rose	Perez Inc
5	John Johnson	76	Male	AB-	Hypertension	2022-08-27	Mary Armstrong	Johnson PLC

insurance_provider character varying (50)	billing_amount numeric (12,2)	room_number integer	admission_type character varying (50)	discharge_date date	medication character varying (100)	test_results character varying (50)
Aetna	18771.01	168	Urgent	2020-09-18	Lipitor	Abnormal
Blue Cross	44582.82	189	Elective	2020-11-30	Penicillin	Abnormal
Aetna	27480.24	253	Urgent	2019-12-24	Aspirin	Normal
Cigna	4407.77	241	Emergency	2023-11-08	Aspirin	Normal
Aetna	21377.00	454	Urgent	2022-09-18	Ibuprofen	Normal

4. Calculating Columns and Measures Using DAX

```
StayLength = DATEDIFF(clean_data[date_of_admission], clean_data[discharge_date], DAY)
```

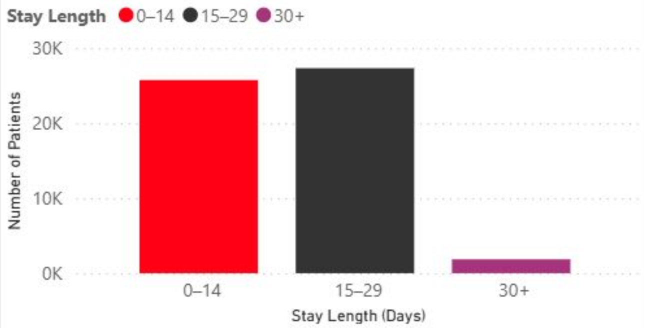
```
Total Billing Amount = SUM(clean_data[billing_amount])
```

```
AvgBilling = AVERAGE(clean_data[billing_amount])
```

Healthcare Analysis

Total Patients	Average Stay Length (Days)	Total Billing Revenue
55K	15.50	1.40bn

Distribution of Patient Length of Stay (Days)

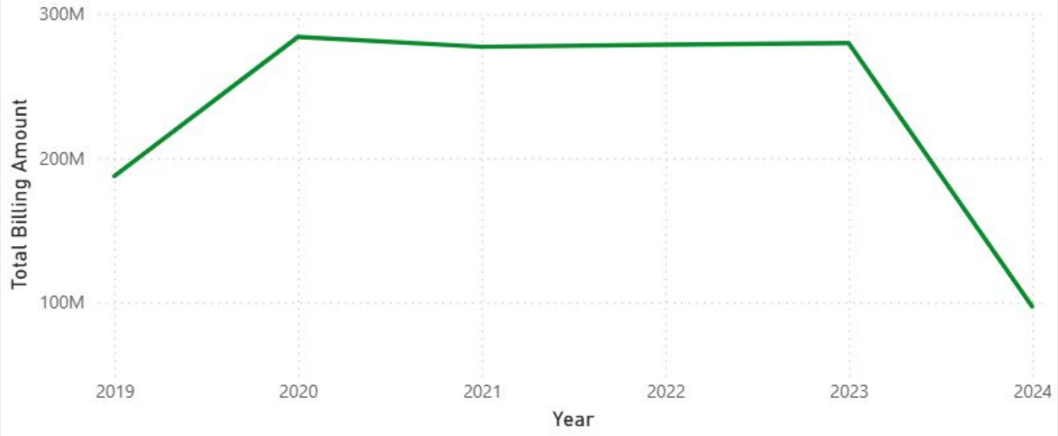


Monthly Patient Admissions Trend

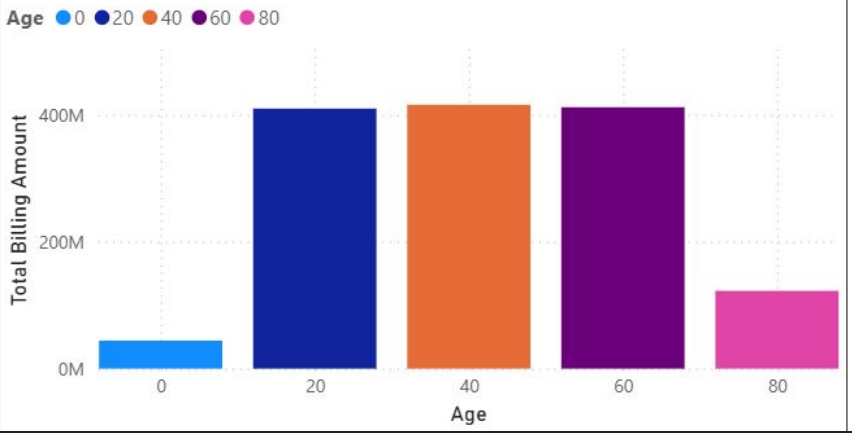
Medical Condition

Arthrit...
Asthma
Cancer
Diabet...
Hyper...
Obesity

Yearly Total Billing Amount



Total Billing amount by Age



6. Key Insights

1. Length of Stay Distribution
   Most patients stay fewer than 30 days; long stays are rare.
   → Consider optimizing care pathways for short-term admissions.

2. Monthly Admissions Trend
   Admissions rise from May to August, possibly due to outdoor activities.
   → Adjust staffing or resources to align with seasonal demand.

3. Yearly Billing Revenue
   Billing was steady from 2020–2023 but dropped by ~$200M in 2024.
   → Explore factors behind the 2024 decline for potential action.

4. Billing by Age Group
   Ages 20–79 drive most billing; under 20 and over 80 contribute less.
   → Consider targeted strategies for younger and older patients.

Conclusion

This analysis provided valuable insights into patient admission patterns, length of stay, and billing trends within a synthetic healthcare dataset. The use of SQL and Power BI enabled effective data cleaning, transformation, and visualization, supporting a deeper understanding of healthcare operations. These findings highlight opportunities for optimizing resource allocation and improving patient care strategies. This project demonstrates the power of data analysis in driving informed decisions in the healthcare sector.